RANDOM FOREST REGRESSION MODEL

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(tidyverse)
library(rio)
library(ggplot2)
library(psych)
library(ggthemes)
library(summarytools)
library(tidyr)
library(formatR)
library(sjmisc)
library(caret)
library(report)
library(knitr)
library(randomForest)
```

Random forest regression:

- is a machine learning algorithm used to predict continous numerical values such as prices, income or temperature
- works by creating many decision tree for anomous prediction

load buiot-in dataset

predict mpg(miles per gallon)

```
data(mtcars)
head(mtcars)
```

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

- the built-in mtcars

check structure

```
str(mtcars)
```

```
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

summary statistics

```
describe(mtcars)
```

```
     vars  n   mean     sd median trimmed    mad   min    max  range
skew
mpg     1 32  20.09   6.03  19.20   19.70   5.41 10.40  33.90  23.50
0.61
cyl     2 32   6.19   1.79   6.00    6.23   2.97  4.00   8.00   4.00 -
0.17
disp    3 32 230.72 123.94 196.30  222.52 140.48 71.10 472.00 400.90
0.38
hp      4 32 146.69  68.56 123.00  141.19  77.10 52.00 335.00 283.00
0.73
drat    5 32   3.60   0.53   3.70    3.58   0.70  2.76   4.93   2.17
0.27
wt      6 32   3.22   0.98   3.33    3.15   0.77  1.51   5.42   3.91
0.42
qsec    7 32  17.85   1.79  17.71   17.83   1.42 14.50  22.90   8.40
0.37
vs      8 32   0.44   0.50   0.00    0.42   0.00  0.00   1.00   1.00
0.24
am      9 32   0.41   0.50   0.00    0.38   0.00  0.00   1.00   1.00
0.36
gear   10 32   3.69   0.74   4.00    3.62   1.48  3.00   5.00   2.00
0.53
carb   11 32   2.81   1.62   2.00    2.65   1.48  1.00   8.00   7.00
1.05
     kurtosis    se
mpg     -0.37  1.07
cyl     -1.76  0.32
disp    -1.21 21.91
hp      -0.14 12.12
drat    -0.71  0.09
wt      -0.02  0.17
qsec     0.34  0.32
vs      -2.00  0.09
am      -1.92  0.09
gear    -1.07  0.13
carb     1.26  0.29
```

check for missing values

```
sum(is.na(mtcars))
```

```
[1] 0
```

```
colSums(is.na(mtcars))
```

```
 mpg  cyl disp   hp drat   wt qsec   vs   am gear carb 
   0    0    0    0    0    0    0    0    0    0    0 
```

split data into train and test

```
set.seed(123)
train_index <- createDataPartition(mtcars$mpg, p = 0.8, list = FALSE)
train_data <- mtcars[train_index, ]
test_data <- mtcars[-train_index, ]
```

check the dimension

```
dim(train_data)
```

```
[1] 28 11
```

```
dim(test_data)
```

```
[1]  4 11
```

Train random forest regression

```
rf_model <- randomForest(mpg ~ ., data = train_data, ntree = 500, mtry
= 3, importance = TRUE)
print(rf_model)
```

```

Call:
 randomForest(formula = mpg ~ ., data = train_data, ntree = 500,
mtry = 3, importance = TRUE)
               Type of random forest: regression
                     Number of trees: 500
No. of variables tried at each split: 3

          Mean of squared residuals: 6.473923
                    % Var explained: 79.44
```

make predictions using testing set

```
predictions <- predict(rf_model, newdata = test_data)
predictions
```

```
 Mazda RX4 Wag     Duster 360 Toyota Corolla Ford Pantera L
      20.78198       14.97612       28.87626       17.19156
```

evaluate model performance

```
rmse <- RMSE(predictions, test_data$mpg)
mae <- MAE(predictions, test_data$mpg)
r2 <- R2(predictions, test_data$mpg)
cat("RMSE", rmse, "\n")
```

```
RMSE 2.630545
```

```
cat("MAE", mae, "\n")
```

```
MAE 1.82736
```

```
cat("R-squared", r2)
```

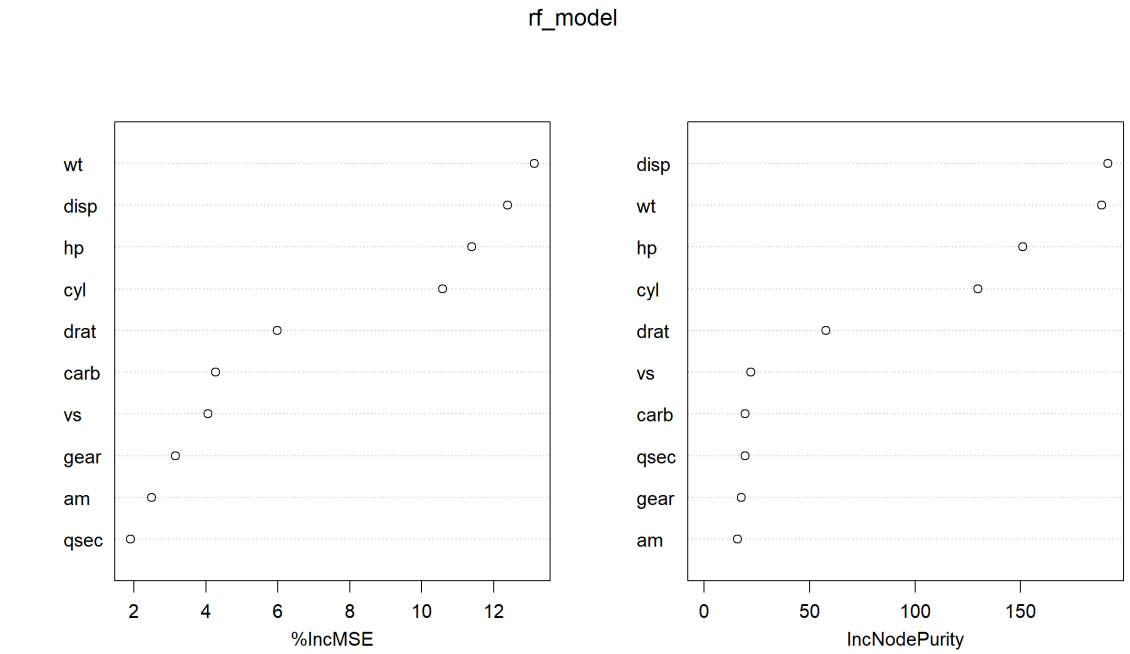
```
R-squared 0.9906742
```

variable importance

```
importance(rf_model)
```

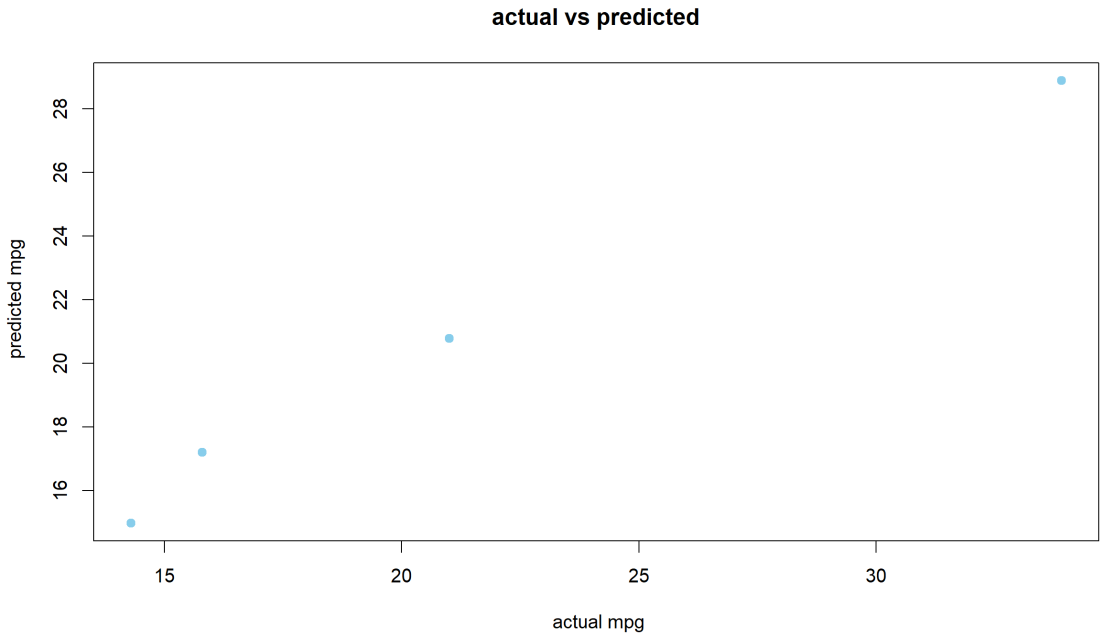
```
       %IncMSE IncNodePurity
cyl  10.582155     129.83235
disp 12.391274     191.37341
hp   11.384711     150.98601
drat  5.986215      57.70490
wt   13.127934     188.44443
qsec  1.908338      19.36439
vs    4.051673      22.08691
am    2.488077      15.89585
gear  3.148357      17.63988
carb  4.263608      19.47617
```

```
varImpPlot(rf_model)
```

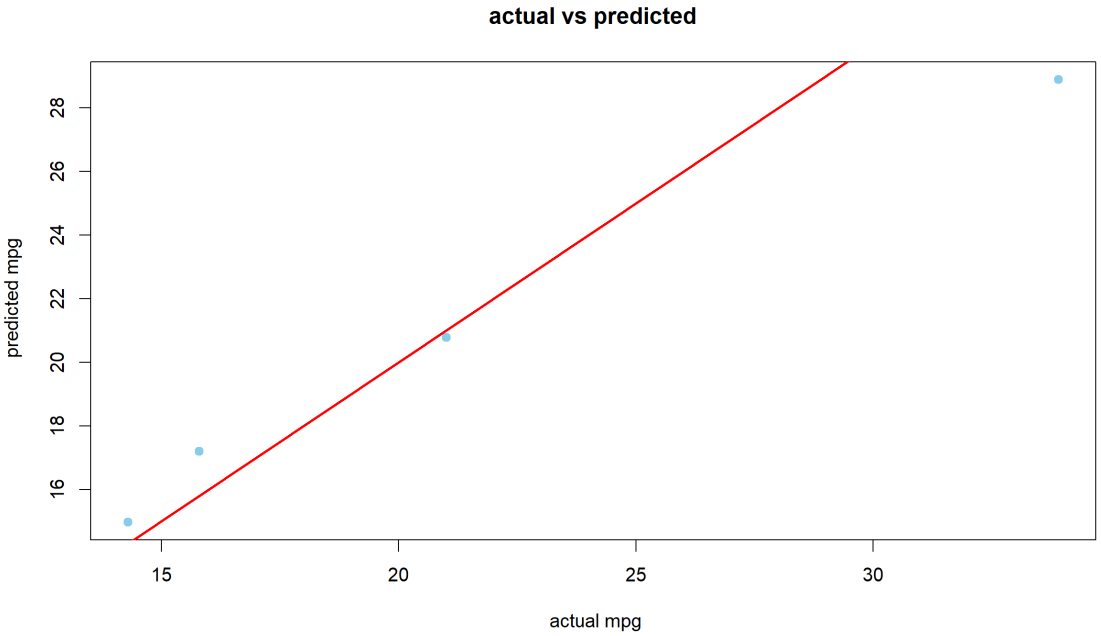


actual vs predicted plot

```
plot(test_data$mpg, predictions, main = "actual vs predicted", xlab =
" actual mpg",
    ylab = " predicted mpg", pch = 19, col = "skyblue")
```




```
abline(0, 1, col = "red", lwd = 2)
```


actual vs predicted
predicted mpg
16
18
20
22
24
26
28
15
20
25
30
actual mpg